IMSC25: Arithmetic functions

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Definition 1. For a positive integer n, denote $\tau(n) = \sum_{d|n} 1$, $\sigma(n) = \sum_{d|n} d$, $\varphi(n) = \sum_{\substack{1 \le k \le n \\ (k,n)=1}} 1$.

All these functions are multiplicative: $(a, b) = 1 \Rightarrow f(a \cdot b) = f(a) \cdot f(b)$.
Recall that $\tau(p^k) = k + 1$, $\sigma(p^k) = \frac{p^{k+1}-1}{p-1}$, $\varphi(p^k) = p^{k-1}(p-1)$.

1 Easier half

1. (Beijing 2022) Prove that for $n \geq 2$, numbers $\varphi(1), \varphi(2), \cdots, \varphi(n)$ can be divided into two groups of equal sum.
2. (AOPS) Assume that for some $n \in \mathbb{Z}^+$, $\log_2 \sigma(n) \in \mathbb{Z}$. Prove that $\log_2 \tau(n) \in \mathbb{Z}$.
3. (AOPS) Find all polynomials $P \in \mathbb{Z}[X]$ for which there exists $k \geq 1$ such that $\varphi(n)|P(n)$ for all integers $n \geqslant k$.
4. (AOPS) Let $M \in \mathbb{Z}^+$ such that for all $n \neq M$, $\varphi(n) \neq \varphi(M)$. Prove that $43 \mid M$.
5. Denote $f(n) := \frac{\sigma(n)}{n}$. Show: $a \mid b \Rightarrow f(a) \leq f(b)$. Prove that f is unbounded. Prove that $\sum_{k=1}^{n} f(k) < 2n$.
6. (MEMO 2024) Find all $P \in \mathbb{Z}[X]$ such that $\sigma(n) \mid P(n)$ for all positive integers n.
7. (SPB 1998) Prove that the sequence $a_n = \tau(n^2 + 1)$ is **not** eventually strictly monotonic.
8. (Bulgaria 2019) Find all $n \in \mathbb{Z}^+$ such that $\varphi(n) \mid n^2 + 3$.
9. (AOPS) Find all monic nonconstant $P \in \mathbb{Z}[X]$ such that $\varphi(P(p)) = P(p-1)$.
10. (Iran 2020) Find all $n \in \mathbb{Z}^+$ for which $\tau(n) \mid 2^{\sigma(n)} - 1$.

2 More difficult half

11. (IMSC NT Mock 2023) Call $n \in \mathbb{Z}^+$ perfect if $\sigma(n) = 2n$. Find all odd $n \in \mathbb{Z}^+$ for which $n^n + 1$ is perfect.
12. Show that for all $n \in \mathbb{Z}^+$ $\tau(2^n - 1) \geq \tau(n)$.
13. (USEMO 2021) Find all integers $n \geq 1$ with $\tau(2^n - 1) = n$.
14. (Iran 2012) Prove that for each $n \in \mathbb{N}$ there exist natural numbers $a_1 < a_2 < ... < a_n$ such that $\varphi(a_1) > \varphi(a_2) > ... > \varphi(a_n)$.
15. Prove or disprove the existence of $a_1, ..., a_{100}$ such that $\tau(a_1 + ... + a_k) = a_k$ for all $k \in \{1, ..., 100\}$.
16. Let $P \in \mathbb{Z}[X]$ with $\deg P \geq 2$. Prove that the sequence $a_n = \tau(P(n))$ is **not** eventually strictly monotonic.
17. (AOPS) Find the greatest $k \in \mathbb{Z}^+$ with the following property: there exists $n \in \mathbb{Z}^+$ such that $\gcd(n + i, \sigma(n + i)) = 1$ for all $i = 1, 2, \cdots, k$.
18. (ELMO 2020) Let $a, b \geq 2$ be integers. Brandon has a calculator with three buttons that replace the integer n currently displayed with $\tau(n)$, $\sigma(n)$, or $\varphi(n)$, respectively. Prove that if the calculator currently displays a, then Brandon can make the calculator display b after a finite (possibly empty) sequence of button presses.
19. (China 2014) Fix $a \geq 9$. Prove that there exist finitely many $n \in \mathbb{Z}^+$ for which $\tau(n) = a$ and $\dfrac{\varphi(n) + \sigma(n)}{n} \in \mathbb{Z}$.
20. (Difficult) Prove that there are infinitely many $n \in \mathbb{Z}^+$ such that $f(n)$ is a perfect square, where a) $f(n) = \sigma(n)$ b) $f(n) = \sigma(n)\varphi(n)$ c) $f(n) = \tau(n)\varphi(n)$.

3 Helpful NT bounds and tools

Proposition 2. 1. $\tau(n) < 2\sqrt{n}$. In general, for each $d > 0$, $\tau(n) < n^d$ for all sufficiently large n.

2. For $n \geq 2$, $n\sqrt{2\tau(n)} \geq \sigma(n) \geq \tau(n)\sqrt{n}$.

3. For nonprime n, $\varphi(n) \leq n - \sqrt{n}$. For $n \geq 7$, $\varphi(n) \geq \sqrt{n}$.

Theorem 3. (Euler's theorem) For $(a, n) = 1$, $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Theorem 4. (Dirichlet's theorem) For $(a, b) = 1$, $|\{ka + b : k \in \mathbb{Z}^+\} \cap \mathbb{P}| = \infty$.

Theorem 5. (Bertrand's postulate) For $n \geq 4$, there exists a prime $p \in (n, 2n)$.

Theorem 6. (CRT) Let $m_1, \ldots, m_k \in \mathbb{Z}^+$ be pairwise coprime. Then the system of congruences

$$\{\forall i \in \{1, 2, \ldots, k\}, \quad x \equiv a_i \pmod{m_i}$$

has a unique solution modulo $M := m_1 m_2 \ldots m_k$, i.e. there is a unique residue $A \in \mathbb{Z}_M$ with $x \equiv A \pmod{M}$.

Theorem 7. (Legendre) $\left(\frac{\cdot}{p}\right) : \mathbb{Z} \to \{-1, 0, 1\}$ finds whether $a \in \mathbb{Z}$ is QR (quadratic residue), NQR or $0 \pmod{p}$. It's useful that $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$ for $p \geq 3$. Moreover, $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$.

(Quadratic reciprocity) For $p, q \in \mathbb{P}$, $p, q \geq 3$, $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}}$.

Theorem 8. (Orders modulo p) For $a \in \mathbb{Z}^+, p \in \mathbb{P}$ with $(a, p) = 1$, we denote by $\operatorname{ord}_p(a)$ the least $k \in \mathbb{Z}^+$ such that $a^k \equiv 1 \pmod{p}$. It's useful that $a^n \equiv 1 \pmod{p} \Rightarrow \operatorname{ord}_p(a) \mid n$. In particular, $\operatorname{ord}_p(a) \mid p - 1$.

Corollary 9. If for $p, q \in \mathbb{P}$ $q \mid 2^p - 1$, then $q > p$.

Theorem 10. (LTE) Let $p \in \mathbb{P}, n \in \mathbb{Z}^+$ and let x, y be integers such that $p \nmid x, y$. Denote by $v_p(n)$ the largest $k \in \mathbb{N}$ for which $p^k \mid n$.

Consider $p \geq 3$. If $p \mid x - y$, then $v_p(x^n - y^n) = v_p(x - y) + v_p(n)$. If $p \mid x + y$ and n is odd, then $v_p(x^n + y^n) = v_p(x + y) + v_p(n)$.

Now consider $p = 2$. If $4 \mid x - y$, then $v_2(x^n - y^n) = v_2(x - y) + v_2(n)$. If $2 \mid x - y$ and n is even, then $v_p(x^n - y^n) = v_p(x^2 - y^2) + v_p(\frac{n}{2}) = v_p(x - y) + v_p(x + y) + v_p(n) - 1$.

Theorem 11. (Zsigmondy's Theorem) Let $a > b \geq 1$ be coprime integers and $n \geq 2$. Then $a^n - b^n$ has a primitive prime divisor p (i.e. $p \nmid a^k - b^k$ for any $1 \leq k < n$), with a few exceptions for $n \in \{1, 2, 6\}$.

The statement also holds about $a^n + b^n$ with one exception for $n = 3$.

Corollary 12. For coprime $a > b \geq 1$ and $n \geq 2$, there exists $p \in \mathbb{P}$, $p \mid a^n - b^n$ with $\operatorname{ord}_p(ab^{-1}) = n$.

Theorem 13. (Mihăilescu's theorem) For $a, b \geq 2$, the only integer solution to $x^a - y^b = 1$ is for $(a, b, x, y) = (2, 3, 3, 2)$.

Proposition 14. The following statements hold:

$$\sum_{k=1}^{\infty} \frac{1}{p_k} \text{ diverges}, \quad \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}, \quad \prod_{p \in \mathbb{P}} \left(1 - \frac{1}{p}\right) = 0.$$

Hints and selected solutions: